



Course Prepared by
IITian & Data Science Architect

PROGRAM DURATION

Total Duration	6 Months
Structured Learning	4 Months
Industry Internship	2 Months

ADVANCED DATA SCIENCE CERTIFICATION PROGRAM

Learn Data. Build Models. Drive Decisions.

SKILL IT Education presents a structured, practical career pathway into Data Science — taking learners step-by-step from mathematics, statistics and Python programming to machine learning, business intelligence and model deployment. Every concept is reinforced through hands-on labs, real-world datasets and industry-aligned data science practices.

PROGRAM HIGHLIGHTS

- ✓ 8 Core Data Science Modules
- ✓ 5+ Hands-On Labs & Capstone Project
- ✓ Machine Learning, BI & Deployment
- ✓ 2-Month Industry Internship

GET IN TOUCH

PHONE

+91 9177715978

EMAIL

info@skilliteducation.com

WEBSITE

www.skilliteducation.com

ADDRESS

LR Towers, No. 3-535, 3rd floor A section, 100 Feet Road, Ayappa Society, Madhapur, Hyderabad, Telangana, India.

YOUR DATA SCIENCE JOURNEY

Math

Python

SQL

EDA

Visualization

BI Tools

ML

Deployment

WHY THIS MATTERS

Organizations across every industry now run on data, and businesses depend on skilled data scientists to clean, analyze, model and deploy insights that drive real decisions — making this one of the fastest-growing and most valuable career paths in technology today.

+91 9177715978 | info@skilliteducation.com | www.skilliteducation.com



Table of Contents

01	Program Overview
02	Module 1 — Mathematics for Data Science
03	Module 2 — Python Programming
04	Module 3 — Data Wrangling (SQL + Cleaning)
05	Module 4 — Exploratory Data Analysis (EDA)
06	Module 5 — Data Visualization (Matplotlib / Seaborn)
07	Module 6 — Business Intelligence Tools (Power BI / Tableau)
08	Module 7 — Machine Learning Fundamentals
09	Module 8 — Model Deployment
10	Data Science Technology Stack
11	Hands-On Labs & Project-Based Learning
12	Learning Outcomes
13	Career Opportunities
14	Why Choose SKILL IT Education

01 Program Overview

Build Your Data Science Career

What This Program Covers

This program takes you step-by-step from strong mathematical and programming foundations to real data science work — cleaning and exploring data, visualizing insights, building machine learning models and deploying them as working applications.

Why This Matters

Every industry now runs on data, and organizations urgently need people who can turn raw data into decisions — not just build reports. Data science skills are among the highest in demand across the global technology job market.

COURSE DURATION

4 Months

PROJECT LEARNING

Integrated

INTERNSHIP

2 Months

PROGRAM ROADMAP

- 1 Mathematics for Data Science
- 2 Python Programming
- 3 Data Wrangling (SQL + Cleaning)
- 4 Exploratory Data Analysis (EDA)
- 5 Data Visualization (Matplotlib / Seaborn)
- 6 Business Intelligence Tools (Power BI / Tableau)
- 7 Machine Learning Fundamentals
- 8 Model Deployment

*From Numbers to Insight. From Models to Decisions.
From Skills to Career.*

02 Module 1 — Mathematics for Data Science

WHAT YOU WILL LEARN

- Linear algebra: vectors & matrices
- Probability & statistics fundamentals
- Descriptive statistics & distributions
- Hypothesis testing & confidence intervals
- Correlation & regression basics
- Calculus for ML: derivatives & gradients
- Combinatorics & probability theory
- Statistical inference for data decisions

TOOLS & TECHNOLOGIES

NumPy

SciPy

Excel

Google Sheets

CORE CONCEPTS

Mean / Median

Variance

Probability

Regression

Distributions

Hypothesis Test

✓ LEARNING OUTCOME

Students build the mathematical and statistical intuition that underpins every data science and machine learning technique used later in the program.

03 Module 2 — Python Programming

WHAT YOU WILL LEARN

- Python syntax, data types & control flow
- Functions, modules & OOP basics
- Lists, dictionaries, tuples & sets
- File handling & exception handling
- NumPy for numerical computing
- Pandas for data manipulation
- Working with virtual environments
- Writing clean, reusable Python code

TOOLS & TECHNOLOGIES

Python

Jupyter Notebook

NumPy

Pandas

VS Code

CORE LIBRARIES

Python

NumPy

Pandas

Jupyter

VS Code

Git

✓ LEARNING OUTCOME

Students gain solid programming fluency in Python, the core language used throughout data science and machine learning.

04 Module 3 — Data Wrangling (SQL + Cleaning)

WHAT YOU WILL LEARN

- Relational databases & SQL fundamentals
- SELECT, JOIN, GROUP BY & subqueries
- Handling missing values & duplicates
- Outlier detection & data standardization
- Merging, reshaping & pivoting datasets
- Feature engineering basics
- Working with APIs & scraped data
- Building clean, analysis-ready datasets

TOOLS & TECHNOLOGIES

SQL

MySQL / PostgreSQL

Pandas

OpenRefine

SQL TOOLKIT

SELECT

JOIN

GROUP BY

Subqueries

Indexes

Views

✓ LEARNING OUTCOME

Students learn to extract, clean and prepare raw data into a reliable, analysis-ready form — the foundation of every downstream data science task.

05 Module 4 — Exploratory Data Analysis (EDA)

WHAT YOU WILL LEARN

- Univariate, bivariate & multivariate analysis
- Summary statistics & data profiling
- Identifying patterns, trends & anomalies
- Correlation analysis between variables
- Outlier detection techniques
- Handling skewed distributions
- Hypothesis-driven data exploration
- Communicating early insights clearly

TOOLS & TECHNOLOGIES

Pandas

NumPy

Jupyter

Pandas Profiling

EDA WORKFLOW

Collect

Profile

Explore

Detect

Summarize

✓ LEARNING OUTCOME

Students learn to systematically explore datasets to uncover patterns and generate the insights that guide every modeling decision that follows.

06 Module 5 — Data Visualization (Matplotlib / Seaborn)

WHAT YOU WILL LEARN

- Principles of effective data visualization
- Line, bar, scatter & histogram plots
- Matplotlib figure & axes customization
- Seaborn statistical plots & heatmaps
- Multi-panel & faceted visualizations
- Storytelling with data
- Choosing the right chart for the data
- Building simple visualization dashboards

TOOLS & TECHNOLOGIES

Matplotlib

Seaborn

Plotly

CHART TYPES

Line

Bar

Scatter

Histogram

Heatmap

Box Plot

✓ LEARNING OUTCOME

Students learn to turn raw analysis into clear, compelling visuals that communicate findings to technical and non-technical audiences alike.

07 Module 6 — Business Intelligence Tools (Power BI / Tableau)

WHAT YOU WILL LEARN

- BI concepts & dashboard design principles
- Connecting & importing data sources
- Building interactive dashboards & reports
- DAX basics in Power BI
- Calculated fields & filters in Tableau
- KPI tracking & drill-down analysis
- Publishing & sharing dashboards
- Presenting insights to stakeholders

TOOLS & TECHNOLOGIES

Power BI

Tableau

Excel

BI WORKFLOW

Connect

Model

Visualize

Publish

Present

✓ LEARNING OUTCOME

Students learn to build and present interactive business dashboards that turn data into decisions for real organizations.

08 Module 7 — Machine Learning Fundamentals

WHAT YOU WILL LEARN

- Supervised vs. unsupervised learning
- Regression: linear & logistic
- Classification: KNN, Decision Trees, Random Forest
- Clustering: K-Means & hierarchical clustering
- Model evaluation: accuracy, precision, recall, F1
- Train-test split, cross-validation & overfitting
- Feature scaling & feature selection
- Introduction to ensemble methods

TOOLS & TECHNOLOGIES

Scikit-learn

Pandas

NumPy

Jupyter

KEY ALGORITHMS

Linear Reg.

Logistic Reg.

KNN

Decision Tree

Random Forest

K-Means

✓ LEARNING OUTCOME

Students learn to build, train and evaluate core machine learning models to solve real prediction and classification problems.

09 Module 8 — Model Deployment

WHAT YOU WILL LEARN

- Model serialization with Pickle / Joblib
- Building REST APIs for ML models
- Deploying models with Flask / FastAPI
- Containerizing applications with Docker
- Deploying models to cloud platforms
- Model monitoring & versioning basics
- Handling real-time predictions
- Building end-to-end ML pipelines

TOOLS & TECHNOLOGIES

Flask

FastAPI

Docker

Streamlit

AWS / Azure Basics

DEPLOYMENT LIFECYCLE

Train

Package

Deploy

Monitor

Update

✓ LEARNING OUTCOME

Students learn to take a trained model out of the notebook and into a working, accessible application — completing the full data science lifecycle.

09 Data Science Technology Stack

The complete toolset used across every module of the program

Programming & Math

- Python
- NumPy / SciPy
- Jupyter Notebook

Data Handling

- SQL
- Pandas
- OpenRefine

Visualization & BI

- Matplotlib / Seaborn
- Power BI
- Tableau

Machine Learning

- Scikit-learn
- Model Evaluation
- Feature Engineering

Deployment

- Flask / FastAPI
- Docker
- Streamlit

Cloud & Versioning

- Git / GitHub
- AWS / Azure Basics
- Model Monitoring

10 Hands-On Labs & Project-Based Learning

Build what you learn — every module is reinforced with hands-on work

01 Data Cleaning Lab

Clean and prepare a messy real-world dataset using SQL and Pandas.

02 EDA & Visualization Project

Explore a dataset end-to-end and present findings through Matplotlib and Seaborn visuals.

03 BI Dashboard Build

Design an interactive Power BI or Tableau dashboard for a real business use case.

04 Machine Learning Model Lab

Train, tune and evaluate classification and regression models on real datasets.

05 Deployment Project

Package a trained model into a working API or app deployed with Flask or Streamlit.

FINAL CAPSTONE

End-to-End Data Science Project

Take a business problem from raw data to a deployed, interactive prediction application.

11 Learning Outcomes

What you will be able to do by the end of the program

- ✓ Apply statistics and probability to real data problems
- ✓ Query and clean data using SQL and Pandas
- ✓ Create clear, effective data visualizations
- ✓ Build and evaluate machine learning models
- ✓ Deploy machine learning models as working applications
- ✓ Communicate data-driven insights to stakeholders
- ✓ Write efficient Python code for data analysis
- ✓ Perform structured exploratory data analysis
- ✓ Build interactive dashboards in Power BI and Tableau
- ✓ Apply feature engineering and model tuning techniques
- ✓ Use Docker and cloud basics for deployment
- ✓ Manage an end-to-end data science project lifecycle

12 Career Opportunities

Build your career in data science and analytics

Data Analysis & BI

- Data Analyst
- Business Intelligence Analyst
- Reporting Analyst

Data Science & ML

- Junior Data Scientist
- ML Engineer (Entry-Level)
- Data Science Associate

Data Engineering

- Junior Data Engineer
- Data Wrangling Specialist
- ETL Analyst

Analytics & Insights

- Analytics Consultant
- Insights Analyst
- Product Analyst

Deployment & MLOps

- ML / MLOps Engineer (Entry)
- Applied Data Scientist

Advanced Career Paths

- Senior Data Scientist
- Data Science Team Lead
- Chief Data Officer (long-term)

Career outcomes depend on individual skills, project execution, interview performance, experience level and market requirements.



WHY CHOOSE SKILL IT EDUCATION

DON'T JUST ANALYZE DATA. LEARN TO ACT ON IT.

Learn. Explore. Model. Deploy. Decide.

SKILL IT Education builds data scientists, not just certificate holders. Through an industry-driven curriculum, hands-on labs and real BI and machine learning tools, every student graduates with practical analytical skills, verified project experience and the confidence to walk into a data science role from day one.

+91 9177715978

info@skilliteducation.com

www.skilliteducation.com

LR Towers, No. 3-535, 3rd floor A section, 100 Feet Road, Ayappa Society, Madhapur, Hyderabad, Telangana, India.

Let's turn data into decisions.

+91 9177715978 | info@skilliteducation.com | www.skilliteducation.com